

Claim Classification Model | TikTok Data Team

Executive Summary



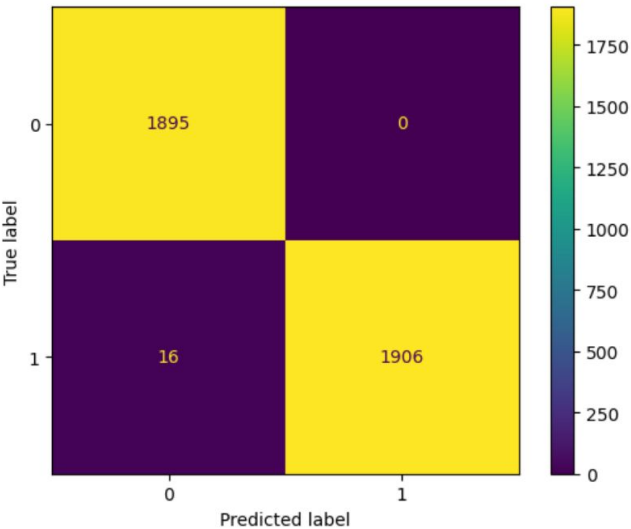
Overview:

At the final stage of developing a machine learning model to classify claim videos. We constructed a random forest model to automate the process of reviewing users’ reports. This summary shows the results and the evaluation of the model with some important insights gained during the process.

Key Insights

- The data was prepared by EDA process and features engineering, then it was splitted into 60% training, 20% validation, 20% and testing sets.
- Two models were developed, XGBoost & Random Forest, where the laser showed better results and selected as the champion.
- The model resulted in a very high score of 99.2% recall, 99.6% f1-score and nearly 100% accuracy and precision.
- The most important features that were founded to determine claim status are interactions records (i.e. count of video views, likes, downloads, shares and comments, respectively.)
- Other features showed negligibly small impact on the model output.

Details



Alt text: Figure shows the confusion matrix of the model.

precision	recall	f1-score
1.0	0.992196	0.996083

Next Steps

- it is highly recommended to start using the model in a free trial because it show good measures and any helpful feedback would be of high benefit.
- Getting the number of reports done by users for each video could be very useful and help us rank claim classified videos based on number of reports and then review them to minimize employee’s effort and save time.